VS-08 Innovation and Methodological Development

Table of contents

1. Purpose and value created . . . 1
2. Position and interaction with the chain . . . 2
3. Stages . . . 2
   - VS-08.1 Identify Innovation Needs and Opportunities . . . 3
   - VS-08.2 Conduct Research and Proof-of-Concepts . . . 3
   - VS-08.3 Evaluate Methodological Soundness and Feasibility . . . 4
   - VS-08.4 Develop and Document New Methodologies or Tools . . . 4
   - VS-08.5 Integrate Innovation into Production Environments . . . 5
   - VS-08.6 Promote Knowledge Sharing and Capability Building . . . 5
4. The four process types . . . 6
   - Design: provides the innovation framework . . . 6
   - Implementation: provides the innovation environment . . . 7
   - Execution: runs the innovation pipeline . . . 8
   - Management: provides innovation performance reports . . . 8
5. Actors and roles . . . 9
6. Capabilities, processes, and functions . . . 10
   - Capability foundations . . . 10
   - Stages, processes, and functions . . . 10
   - Framework mappings . . . 11
7. Principles . . . 11
8. Interfaces, services, and events . . . 12
9. Key performance indicators . . . 13
10. Toolkit . . . 13

1. Purpose and value created

VS-08 makes sure that the NSI does not stand still. It discovers, tests, and introduces new ways of working: new data sources, methods, tools, and technologies. It connects research, methodology, IT, and business users. It keeps innovation controlled, so that only methods that are sound, documented, and compliant reach production. VS-08 does not produce statistics. It improves the methods and tools that the production value streams use, and it builds the skills needed to apply them. The value created is **continuous improvement**: the NSI stays ready for new data, such as sensor data, administrative data, and data spaces, and for new techniques, such as machine learning and small-area estimation.

VS-08 is a supporting value stream. It aligns with the GAMSO area *Capability Management* rather than with the production phases of GSBPM. This section describes its business layer for all four process types.

- **Methodologist, researcher, or data scientist:** the stages in attribute 3, which describe the innovation pipeline.
- **Head of methodology or innovation:** the Design processes in attribute 4.
- **Process owners of the production value streams:** attributes 2 and 8, which describe how needs enter VS-08 and how validated methods return.

- **Manager who wants the big picture:** attributes 1, 2, and 9. Examples and scenarios are part of the SAF Toolkit.

> **How a supporting value stream differs from a primary one**
>
> VS-08 has no steady-state data transition. It does not move data from one state to the next; it improves the methods and tools that the production value streams use. It runs continuously as a pipeline of research and development, not per production cycle. Its output is an enabling capability rather than a data artifact: validated methods, documented tools, reference implementations, and trained staff. Where it uses production data for research, it does so under specific authorization and within the rules of VS-07.

PLACEHOLDER: ArchiMate view VS-08 (a) value stream and capabilities

2. Position and interaction with the chain

VS-08 is not a stage in the production line. It runs alongside the chain as a pipeline that feeds improvements into every primary value stream and draws its agenda from their needs. The improvement recommendations of the primary value streams, produced in their *Continuous Improvement* sub-processes, together with the user needs from VS-10, form the innovation backlog of VS-08. After research and evaluation, the transfer gate in VS-08.5 moves a validated method or tool into the production value stream concerned. That transfer requires the approval of VS-07 and runs on the platforms of VS-09.

Counterpart	VS-08 provides	VS-08 receives
VS-01 to VS-06 (primary value streams)	Validated methods and tools transferred to production	Improvement recommendations, problems with current methods, validation of fitness for purpose
VS-07 Governance and Compliance	Methods submitted for approval before production	Compliance criteria and approval decisions
VS-09 Infrastructure and Platform Enablement	Requirements for research and production platforms	Research environments (sandboxes) and production runtime
VS-10 Customer and Stakeholder Engagement	New possibilities for products and services	User needs that require new methods or sources
External parties: universities, other NSIs, international bodies	Shared methods and open-source contributions	Research results, shared methods, and tools

Occasional access to data in the steady states is part of this interaction. Researchers in VS-08 may explore Standardized Data (#2) or Processed Data (#3) to test new methods, under specific authorization (see *State #2: Standardized Data* and *State #3: Processed Data*).

PLACEHOLDER: ArchiMate view VS-08 (c) interaction with the chain: improvement needs from VS-01 to VS-06 and user needs from VS-10 entering the innovation backlog; the pipeline of research, evaluation, and development; the transfer gate to production with approval of VS-07 and platforms of VS-09

3. Stages

The stages of VS-08 form a continuous innovation pipeline: identify, research, evaluate, develop, integrate, and share. Ideas enter at the start, and many of them are stopped along the way. That is intended: a healthy pipeline stops attractive but unsound ideas before they reach production.

Stage	Name	Value added
VS-08.1	Identify Innovation Needs and Opportunities	It is clear which innovations are worth pursuing
VS-08.2	Conduct Research and Proof-of-Concepts	A new method or technology has been tested in a limited scope
VS-08.3	Evaluate Methodological Soundness and Feasibility	Only sound and feasible innovations go forward
VS-08.4	Develop and Document New Methodologies or Tools	A validated method is reproducible and ready to hand over
VS-08.5	Integrate Innovation into Production Environments	The method or tool works as part of a production value stream
VS-08.6	Promote Knowledge Sharing and Capability Building	The organization can adopt and sustain the new way of working

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. Research has no reference process in the SAF; it is carried out through innovation projects, which the column shows as *NSI-specific*. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-08.1 Identify Innovation Needs and Opportunities

Value created: *A prioritized backlog of innovations that are worth pursuing.*

The pipeline begins by collecting needs and opportunities. The improvement recommendations of the production value streams and the user needs from VS-10 enter the backlog. The team scans technological, methodological, and policy developments, such as artificial intelligence, new data sources, and cloud services. It checks whether a source or method that already exists elsewhere, in the NSI or at another NSI, can meet the need. Each candidate is assessed for value to the organization, feasibility, and risk. The result is a prioritized innovation backlog, aligned with the strategy of the NSI.

Activity	Key outputs	SAF process	Other reference
Collect improvement needs from the value streams and user needs from VS-10 Customer and Stakeholder Engagement	Raw backlog	NSI-specific	4.1.4 Improvement management
Scan technological, methodological, and policy developments	Horizon scan	NSI-specific	2.2.2 Product innovation
Check whether existing sources or methods can meet the need	Availability check	PRD-1.5 Check data availability and suitability	2.2.1 Data source exploration; SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Assess and prioritize candidates by value, feasibility, and risk	Prioritized innovation backlog	NSI-specific	SAF-P04 Maximize the benefits for the organization

VS-08.2 Conduct Research and Proof-of-Concepts

Value created: *A new method or technology tested in a limited, controlled scope before any commitment.*

Candidates from the backlog enter research. The team defines a research question, a hypothesis, and criteria for success before it starts. It runs controlled experiments and pilots in a research environment provided by VS-09. Where production data is needed, access is requested under specific authorization, and the data is used only for the stated purpose. Results are recorded, including negative results, so that the organization learns from what did not work. A proof-of-concept ends with a clear recommendation: stop, adjust, or proceed to evaluation.

Activity	Key outputs	SAF process	Other reference
Define the research question, hypothesis, and success criteria	Research plan	NSI-specific	2.2.3 Methodology
Run experiments and pilots in a research environment	Experiment and pilot results	NSI-specific	2.2.1 Data source exploration
Request authorized access to production data where needed	Access authorization	DLM-6 Provide data for reuse	SAF-P11 Privacy by design
Record results, including negative results, and recommend a next step	Proof-of-concept report	NSI-specific	

VS-08.3 Evaluate Methodological Soundness and Feasibility

Value created: *Confidence that only sound, feasible, and compliant innovations go forward.*

Promising results are evaluated before any further investment. The evaluation assesses accuracy, bias, robustness, and scalability. It checks alignment with the quality framework and with ethical rules, for example on transparency and explainability of machine-learning methods. It compares the new method with the current method on real or realistic data. Subject-matter experts from the production value stream concerned confirm fitness for purpose. VS-07 assesses the method against its compliance criteria. This gate stops ideas that are attractive but not sound, with a documented rationale.

Activity	Key outputs	SAF process	Other reference
Assess accuracy, bias, robustness, and scalability	Soundness assessment	PRD-3.6 Test statistical business process	2.2.3 Methodology
Compare the new method with the current method on realistic data	Comparison report	PRD-3.6 Test statistical business process	GAMSO Quality Management
Confirm fitness for purpose with the production value stream	Fitness-for-purpose confirmation	PRD-3.6 Test statistical business process	SAF-P05 IT and technology is everyone's business
Obtain the compliance assessment of VS-07 Governance and Compliance	Compliance assessment	NSI-specific	SAF-P11 Privacy by design

VS-08.4 Develop and Document New Methodologies or Tools

Value created: *A validated method that is reproducible and ready to hand over, rather than living in a researcher's notebook.*

What passes the gate is developed into a standard method, algorithm, or reference implementation. The method is documented: its purpose, assumptions, parameters, limitations, and the evidence from the evaluation. The reference implementation is written so that it can be maintained, tested, and run on the

production platforms of VS-09. Where possible, it is developed or shared as open source with other NSIs. The method is added to the knowledge base, so that others can find and apply it.

Activity	Key outputs	SAF process	Other reference
Develop the standard method, algorithm, or reference implementation	Reference implementation	PRD-3.2 Reuse or build processing and analysis components	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Document purpose, assumptions, parameters, limitations, and evidence	Method documentation	OVA-5 Knowledge Management	SAF-P09 No data without metadata and classification
Prepare the implementation to run on production platforms	Production-ready component	PRD-3.2 Reuse or build processing and analysis components	SAF-P07 Implement loosely coupled processes and systems
Add the method to the knowledge base	Knowledge base entry	OVA-5 Knowledge Management	2.2.3 Methodology

VS-08.5 Integrate Innovation into Production Environments

Value created: *A validated method or tool working as part of a production value stream.*

This is the transfer gate, where innovation becomes part of everyday production. The method or tool is handed over to the production value stream concerned. That value stream takes it up through its own Design and Implementation process types. Approval from VS-07 is confirmed before the transfer. The component is deployed on the platforms of VS-09 and tested in the production context. Ownership is transferred formally, so that the production value stream maintains the method from then on. The transfer is recorded, together with the method version and the date of entry into production.

Activity	Key outputs	SAF process	Other reference
Hand over the method to the Design and Implementation of the production value stream	Handover package	PRD-3.7 Finalize production systems	4.1.3 Change management
Confirm the approval of VS-07 Governance and Compliance	Approval record	NSI-specific	4.1.5 Accountability
Deploy and test on the platforms of VS-09 Infrastructure and Platform Enablement	Production deployment	PRD-3.7 Finalize production systems	3.2.3 Technical management
Transfer ownership formally and record the transfer	Transfer record, *Method transferred to production*	PRD-3.7 Finalize production systems	SAF-P03 Active life cycle management

VS-08.6 Promote Knowledge Sharing and Capability Building

Value created: *An organization that can adopt and sustain the new way of working.*

The pipeline ends by spreading what was learned. Results, best practices, and guidelines are shared inside the NSI and with the international community. Staff in the production value streams are trained in the new method. Communities of practice keep knowledge alive after the project ends. Superseded methods are retired formally. The experience gained surfaces the next round of needs, which returns to VS-08.1.

Activity	Key outputs	SAF process	Other reference
Share results, best practices, and guidelines inside and outside the NSI	Published guidelines	OVA-5 Knowledge Management	4.1.6 Collaboration
Train staff in the production value streams	Trained staff	OVA-5 Knowledge Management	3.1.1 Personnel management (HR)
Support communities of practice	Active communities of practice	OVA-5 Knowledge Management	SAF-P05 IT and technology is everyone's business
Retire superseded methods	Retirement record	OVA-5 Knowledge Management	SAF-P03 Active life cycle management

4. The four process types

VS-08 is itself a capability that must be designed, built, operated, and improved. The four process types therefore apply to VS-08 as they do to the primary value streams, but they govern the innovation capability rather than a data pipeline. Design produces the innovation strategy and the evaluation criteria, Implementation builds the research environment and the pipeline procedures, Execution runs the stages of attribute 3, and Management evaluates whether innovation delivers. **[VC26]**

Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. Where no reference process exists, the sub-process is marked as NSI-specific.

Design: provides the innovation framework

Design sub-process	Refines	Main outputs
Design the Innovation Strategy and Portfolio	NSI-specific	Innovation strategy, portfolio themes, budget rules
Design the Evaluation and Transfer Criteria	PRD-2.5 Design processing and analysis; OVA-1 Quality Management	Evaluation criteria, gate criteria, transfer protocol

Design the Innovation Strategy and Portfolio

Refines: NSI-specific strategic planning processes.

This sub-process sets the direction of innovation. It derives innovation themes from the strategy of the NSI, such as the use of new data sources or the automation of coding. It decides how the innovation portfolio is balanced between quick improvements and long-term research. It sets the rules for funding and prioritizing projects. It defines how the NSI cooperates with universities, other NSIs, and international bodies. The strategy keeps innovation focused on value for the organization, in line with SAF-P04, rather than on novelty.

Activity	Key outputs	SAF process	Other reference
Derive innovation themes from the strategy of the NSI	Innovation strategy	NSI-specific	4.1.1 Strategy and governance
Define the balance of the portfolio and the rules for funding and priority	Portfolio rules	NSI-specific	SAF-P04 Maximize the benefits for the organization
Define cooperation with universities, other NSIs, and international bodies	Cooperation framework	NSI-specific	4.1.6 Collaboration

Design the Evaluation and Transfer Criteria

Refines: PRD-2.5 Design processing and analysis; OVA-1 Quality Management.

This sub-process designs the gates of the pipeline. It sets the criteria for soundness: accuracy, bias, robustness, scalability, and explainability. It sets the criteria that a method must meet before transfer, together with VS-07. It designs the transfer protocol: what must be handed over, to whom, and with which documentation. It defines when a method is retired. Clear criteria make gate decisions predictable and defensible.

Activity	Key outputs	SAF process	Other reference
Define criteria for methodological soundness	Evaluation criteria	PRD-2.5 Design processing and analysis	2.2.3 Methodology
Define the criteria for transfer, together with VS-07 Governance and Compliance	Gate criteria	OVA-1 Quality Management	4.1.5 Accountability
Design the transfer protocol and the rules for retirement	Transfer protocol	NSI-specific	SAF-P03 Active life cycle management

Implementation: provides the innovation environment

Implementation sub-process	Refines	Main outputs
Set Up Research Environments and Procedures	PRD-3.4 Configure workflows; DLM-6 Provide data for reuse	Research environment, access procedure for research data, pipeline procedures
Build the Knowledge Base	OVA-5 Knowledge Management	Knowledge base, method catalog, training material

Set Up Research Environments and Procedures

Refines: PRD-3.4 Configure workflows; DLM-6 Provide data for reuse.

This sub-process builds the environment in which research takes place, together with VS-09. It provides research environments, or sandboxes, with the tools that researchers need. It defines the procedure for authorized access to production data for research, with VS-07. It sets up the procedures for the pipeline: how ideas are submitted, how gate decisions are taken, and how results are recorded. It makes sure that experiments can be reproduced. A good environment lets researchers move fast without putting production or confidentiality at risk.

Activity	Key outputs	SAF process	Other reference
Provide research environments with VS-09 Infrastructure and Platform Enablement	Research environment	PRD-3.4 Configure workflows	3.2.5 IT infrastructure
Define the procedure for authorized access to production data	Access procedure for research data	DLM-6 Provide data for reuse	SAF-P11 Privacy by design
Set up the pipeline procedures: submission, gate decisions, recording	Pipeline procedures	NSI-specific	4.1.4 Improvement management

Build the Knowledge Base

Refines: OVA-5 Knowledge Management.

This sub-process builds the place where methods and knowledge are kept. It sets up the method catalog with the documentation of validated methods and their versions. It links each method to the production value streams that use it. It develops training material and learning paths for new methods. It provides the channels for communities of practice. The knowledge base turns individual expertise into an asset of the organization.

Activity	Key outputs	SAF process	Other reference
Set up the method catalog with documentation and versions	Method catalog	OVA-5 Knowledge Management	2.2.3 Methodology
Link methods to the value streams that use them	Method usage map	OVA-5 Knowledge Management	SAF-P09 No data without metadata and classification
Develop training material and support for communities of practice	Training material, community channels	OVA-5 Knowledge Management	3.1.1 Personnel management (HR)

Execution: runs the innovation pipeline

Execution runs the stages of attribute 3 continuously, as a pipeline of projects. VS-08.1 runs as a periodic planning cycle; VS-08.2 to VS-08.5 run per project; VS-08.6 runs continuously.

Aspect	VS-08
Trigger	*Innovation need raised* by a value stream or VS-10 Customer and Stakeholder Engagement; a planning cycle; a new opportunity from the horizon scan
Cadence	Per project; periodic review of the backlog
Performed by	ROL-07 Methodologist, ROL-15 Subject-matter statistician (researcher, data scientist), ROL-10 Data engineer
Input	Improvement needs, user needs, research results from outside, authorized research data
Output	Validated methods and tools, documentation, transfer records, trained staff
Control point	Gate in VS-08.3 Evaluate Methodological Soundness and Feasibility; transfer gate in VS-08.5 Integrate Innovation into Production Environments
Exception path	An unsound method stopped with a documented rationale; a compliance objection returned to research or stopped

Management: provides innovation performance reports

Management sub-process	Refines	Main outputs
Monitor the Innovation Pipeline	PRD-8.1 Gather evaluation inputs	Pipeline dashboard, lead times, gate statistics
Evaluate and Improve the Innovation Capability	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Evaluation of transfers, improvement plan

Monitor the Innovation Pipeline

Refines: PRD-8.1 Gather evaluation inputs.

This sub-process tracks how the pipeline performs. It measures the key performance indicators of VS-08 (see attribute 9), such as the time from idea to production and the share of proofs-of-concept stopped at the gate. It tracks the status of each item in the backlog. It records the use of research data and the compliance of research projects. It reports the progress of the portfolio to management. Visibility of the pipeline helps the value streams see when their needs will be addressed.

Activity	Key outputs	SAF process	Other reference
Measure the key performance indicators of VS-08 Innovation and Methodological Development	Pipeline dashboard	PRD-8.1 Gather evaluation inputs	GAMSO Capability Management
Track the status of backlog items	Backlog status report	PRD-8.1 Gather evaluation inputs	
Record the use of research data and the compliance of research projects	Research compliance record	DLM-8 Monitor data quality, lineage and use	SAF-P11 Privacy by design

Evaluate and Improve the Innovation Capability

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This sub-process asks whether innovation delivers lasting value. It evaluates transferred methods after a period in production: are they still in use, and did they deliver the expected benefits? It holds retrospectives on stopped and completed projects. It adjusts the strategy, the evaluation criteria, and the procedures based on the findings. It prioritizes improvements of the innovation capability and tracks them. This closes the cycle of VS-08 itself, in line with SAF capability 4.1.4 *Improvement management*.

Activity	Key outputs	SAF process	Other reference
Evaluate transferred methods after a period in production	Transfer evaluation	PRD-8.2 Conduct evaluation	GAMSO Capability Management
Hold retrospectives on stopped and completed projects	Lessons learned	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Adjust strategy, criteria, and procedures, and track the improvements	Improvement plan	PRD-8.3 Agree an action plan	4.1.4 Improvement management

5. Actors and roles

VS-08 connects researchers, methodologists, IT staff, and the production value streams. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the head of methodology or innovation is ROL-07 Methodologist, the researcher or data scientist is ROL-15 Subject-matter statistician, and the tool or platform engineer is ROL-10 Data engineer. The trainer or knowledge manager is not part of the reference set. **[VC28]**

Actor or role	ID	Responsibility in VS-08 Innovation and Methodological Development
International statistical organization	ACT-07	Shares methods, standards, and research results, for example through UNECE projects
Supplier and service provider	ACT-09	Supplies research partnerships, tools, or data for innovation, for example universities
Process owners of production value streams	ROL-01	Raise needs, confirm fitness for purpose, and take over transferred methods
Methodologist	ROL-07	Owns the innovation agenda and the methodological standards; evaluates soundness and formalizes validated methods
Data engineer	ROL-10	Builds reference implementations and prepares them for production, together with VS-09 Infrastructure and Platform Enablement
Subject-matter statistician	ROL-15	Carries out research and proofs-of-concept; brings the real problem from production and validates fitness for purpose
Compliance officer	ROL-19	Assesses methods against the compliance criteria of VS-07 Governance and Compliance before transfer

6. Capabilities, processes, and functions

Capability foundations

Capability group	Capabilities	Role in VS-08 Innovation and Methodological Development
2.2 Research	2.2.1 Data source exploration, 2.2.2 Product innovation, 2.2.3 Methodology	Explore new sources, develop new products and methods
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Bring validated methods into the design and build of products
3.2 Computerization and automation	3.2.1 Application development, 3.2.3 Technical management	Build reference implementations and deploy them
4.1 Strategy and leadership	4.1.3 Change management, 4.1.4 Improvement management, 4.1.6 Collaboration	Manage the change that innovation brings, improve continuously, cooperate with partners

Stages, processes, and functions

The table follows the mapping in *Business Processes, Mapping to value stream stages.*

Stage	Realized by processes	Main functions
VS-08.1 Identify Innovation Needs and Opportunities	PRD-1.5 Check data availability and suitability	FUN-4.2 Research and innovate
VS-08.2 Conduct Research and Proof-of-Concepts	No reference process (NSI-specific)	FUN-4.2 Research and innovate
VS-08.3 Evaluate Methodological Soundness and Feasibility	PRD-3.6 Test statistical business process	FUN-4.1 Manage statistical and analytical methodology; FUN-5.2 Assure and review quality
VS-08.4 Develop and Document New Methodologies or Tools	PRD-3.2 Reuse or build processing and analysis components; OVA-5 Knowledge Management	FUN-4.1 Manage statistical and analytical methodology; FUN-10.4 Manage information and records
VS-08.5 Integrate Innovation into Production Environments	PRD-3.7 Finalize production systems	FUN-2.4 Transfer capability improvements to operations; FUN-3.1 Design and build statistical products
VS-08.6 Promote Knowledge Sharing and Capability Building	OVA-5 Knowledge Management	FUN-10.1 Manage human resources; FUN-10.4 Manage information and records

PLACEHOLDER: ArchiMate view VS-08 (b) stages realized by processes and functions

Framework mappings

Where the primary value streams map to the production phases of GSBPM, VS-08 maps to GAMSO *Capability Management*.

GAMSO area	Relevance to VS-08 Innovation and Methodological Development	SAF capabilities
Capability Management	Manage research and innovation, develop methodology, manage change, build staff capability	2.2.1 Data source exploration, 2.2.2 Product innovation, 2.2.3 Methodology, 4.1.3 Change management
Strategy and Leadership	Set the innovation agenda in line with the strategy of the organization	4.1.1 Strategy and governance
Corporate Support	Train staff and manage knowledge	3.1.1 Personnel management (HR), 3.1.8 Information management and archive management

7. Principles

Principle	Statement	How VS-08 Innovation and Methodological Development applies it
SAF-P01	Reuse before open source, open source before buying, buying before building	Innovates on and contributes to shared, open methods and tools rather than closed products from vendors [**VC3**]

Principle	Statement	How VS-08 Innovation and Methodological Development applies it
SAF-P03	Active life cycle management	Manages methods and tools across their life cycle, from proof-of-concept to production to retirement
SAF-P04	Maximize the benefits for the organization	Prioritizes innovations by value for the whole organization and by scalability, not by novelty
SAF-P05	IT and technology is everyone's business	Connects research, methodology, IT, and business users, and builds broad capability to adopt new methods
SAF-P07	Implement loosely coupled processes and systems	Transfers methods that fit the steady-state interfaces, so that the production value streams stay decoupled

8. Interfaces, services, and events

Business services

Service	Offered to	Realized in	Interfaces
Innovation intake service	All value streams, VS-10 Customer and Stakeholder Engagement	VS-08.1 Identify Innovation Needs and Opportunities	Innovation backlog, submission procedure
Research and experimentation service	Production value streams	VS-08.2 Conduct Research and Proof-of-Concepts	Research environments, proof-of-concept reports
Methodology advice and evaluation service	Production value streams, VS-07 Governance and Compliance	VS-08.3 Evaluate Methodological Soundness and Feasibility	Soundness assessments, comparison reports
Method transfer service	Production value streams	VS-08.5 Integrate Innovation into Production Environments	Transfer protocol, handover packages
Knowledge and training service	All staff	VS-08.6 Promote Knowledge Sharing and Capability Building	Method catalog, training, communities of practice

Business events

Event	Triggers or marks
Innovation need raised	Entry of an item into the backlog in VS-08.1 Identify Innovation Needs and Opportunities
Proof-of-concept started / *Proof-of-concept completed*	Start and end of VS-08.2 Conduct Research and Proof-of-Concepts
Method compliance assessment requested	Request to VS-07 Governance and Compliance during VS-08.3 Evaluate Methodological Soundness and Feasibility
Gate passed / *Gate stopped*	Outcome of VS-08.3 Evaluate Methodological Soundness and Feasibility

Event	Triggers or marks
Platform request	Request to VS-09 Infrastructure and Platform Enablement for a research or production environment
Method transferred to production	Completion of VS-08.5 Integrate Innovation into Production Environments
Method retired	Formal retirement of a superseded method in VS-08.6 Promote Knowledge Sharing and Capability Building

PLACEHOLDER: VS-08 event choreography between the production value streams, VS-08, VS-07, and VS-09: need raised, proof-of-concept, compliance assessment, gate decision, platform request, transfer to production, and retirement

9. Key performance indicators

KPI	Description	Measurement
Idea-to-production lead time	Median time from a backlog item in VS-08.1 Identify Innovation Needs and Opportunities to a completed transfer in VS-08.5 Integrate Innovation into Production Environments	Per transferred innovation
Transfer rate and durability	Methods and tools transferred per year, and the share still in use after 12 months	Annually
Gate effectiveness	Share of proofs-of-concept stopped at the gate of VS-08.3 Evaluate Methodological Soundness and Feasibility with a documented rationale; a healthy gate stops some	Per evaluation
Backlog responsiveness	Share of prioritized backlog items that reach a proof-of-concept within the agreed time	Per planning period
Capability building	Staff trained in new methods, and the rate of adoption in the target value streams	Per year, per transfer

10. Toolkit

Examples from NSI practice and a maturity checklist for VS-08 are part of the SAF Toolkit.

Toolkit: SAF-TK-V08 VS-08 Innovation and Methodological Development companion **[VC17]**